

Deploying

Where this can run

The Claude Agent SDK spawns a Claude Code subprocess. That one fact decides the host.

	RENDER / RAILWAY / FLY	VERCEL
Agent subprocess	Works	Does not work
Size limit	None that matters	250MB unzipped per function
Subprocess lifetime	Survives between requests	Re-spawned on every cold start
Good for	The whole app	Nothing here

Vercel cannot host this app. It is worth stating plainly because the build succeeds and the failure only shows up as the agent saying "I'm having trouble processing that right now" on every turn:

- The SDK's per-platform native CLI is about **238MB**. A Vercel function is capped at 250MB unzipped, and Next plus the rest of `node_modules` does not fit in what is left. The function deploys, then every turn fails with `Native CLI binary for linux-x64 not found`.
- The SDK also ships a smaller `cli.js` that does fit, and pointing `pathToClaudeCodeExecutable` at it was tried. The agent starts, but **MCP tools never attach**: it loops on `ToolSearch` and never finds `lookup_transaction`, so it cannot answer anything factual. Reproduced on consecutive runs.

There is no configuration that resolves both, so the app is deployed to a persistent Node host.

Render

`render.yaml` in the repo is a Blueprint. In Render: **New** → **Blueprint**, point it at the repo, and it reads that file.

SETTING	VALUE	WHY
Runtime	Node	
Build Command	<code>npm install && npm run build</code>	builds the workspace packages, then Next
Start Command	<code>npm start</code>	<code>next start</code> , which honours Render's <code>PORT</code>
Health Check Path	<code>/api/vapi</code>	returns a small JSON payload
<code>NODE_VERSION</code>	22	Node 22 has a native WebSocket, which <code>supabase-js</code> requires

PLAN

The **free** plan spins the service down after about 15 minutes idle, and the next request pays a cold start *plus* the agent subprocess spawn. On a voice call that is the caller waiting through silence. **Starter** keeps it running, which also means the warm-session work survives between calls rather than only within one.

ENVIRONMENT VARIABLES

Set these in the Render dashboard (the Blueprint declares them `sync: false` , so Render prompts rather than storing them in git):

VARIABLE	NOTES
ANTHROPICAPIKEY	
SUPABASE_URL	Bare project URL, no <code>/rest/v1</code>
SUPABASESERVICEROLE_KEY	Server-only. Never prefix with <code>NEXTPUBLIC</code> .
AUTH_SECRET	32+ random chars. Changing it logs everyone out.
ADMINEMAIL , ADMIN PASSWORD	There is no sign-up; this is the only account
VAPISERVERSECRET	Must match the Vapi assistant's secret
NEXTPUBLICVAPIPUBLICKEY	Publishable; safe in the browser
NEXTPUBLICVAPIASSISTANTID	Publishable
RESENDAPIKEY , EMAIL_FROM	<code>EMAIL_FROM</code> must be on a verified Resend domain
ALLOWEDORIGINS , APP BASE_URL	Your Render URL
VOYAGEAPIKEY or OPENAIAPIKEY	Optional; without either, retrieval uses Postgres full-text
CLAUDE_MODEL	Optional; defaults to <code>claude-haiku-4-5</code> . If you set this to Sonnet it overrides the default and turns get slower — warm turns averaged 14.6s on Sonnet against 9.2s on Haiku.
AGENTTURNTIMEOUT_MS	Optional; defaults to 90s

**** `NEXTPUBLIC*` variables are baked in at build time.**** Adding one after a deploy does nothing until you redeploy. If the call page says "Voice isn't configured" where the variable is clearly set, this is why.

POINT VAPI AT IT

See [vapi-configuration.md](#). In short:

- **Model** → **Custom LLM URL:** `https://<your-service>.onrender.com/api/vapi`
- **Server URL:** `https://<your-service>.onrender.com/api/vapi/end-of-call`
- Both secrets equal to `VAPISERVERSECRET`

Local development

Next only auto-loads `.env` from its own directory (`apps/web`), so every env-dependent npm script runs through `scripts/with-env.mjs` , which loads the root file into the environment *before* Next starts. That ordering matters: loading it from `next.config.mjs` is too late, because Next restores its own snapshot of `process.env` afterwards and edge middleware cannot read files at all.

```
npm install
cp .env.example .env      # fill it in
npm run build              # workspace packages must be built before the agent can spawn the MCP server
npm run dev
```

EXPOSING LOCALHOST TO VAPI

Vapi calls your server from its cloud, so `localhost` is unreachable:

```
npx ngrok http 3000
```

Point the assistant's Custom LLM URL at `https://<id>.ngrok.app/api/vapi` . The free tier changes the subdomain on every restart.

Troubleshooting

SYMPTOM	CAUSE
Every turn says "I'm having trouble processing that right now"	The agent ran but failed. The response carries <code>x-relaypay-degraded</code> with the reason — read it with curl, it is only visible to a caller holding <code>VAPISERVERSECRET</code> .
<code>Native CLI binary for linux-x64 not found</code>	You are on Vercel, or a host that omitted optional dependencies. See the top of this page.
Agent replies but calls no tools, looping on <code>ToolSearch</code>	MCP never attached. Check <code>packages/mcp-server/dist</code> exists (<code>npm run build</code>).
"Voice isn't configured" despite the env being set	<code>NEXTPUBLIC*</code> is build-time. Redeploy. Locally, check the <code>.env</code> is at the repo root .
Build fails on <code>@relaypay/shared</code> not found	The workspace packages were not compiled. <code>apps/web</code> 's <code>prebuild</code> hook does this.
Vapi gets 401	<code>VAPISERVERSECRET</code> differs between the deployment and the assistant settings.
Emails fail with a domain error	<code>EMAIL_FROM</code> is not on a domain verified in Resend.
Admin login returns 500 <code>not_configured</code>	<code>ADMINEMAIL</code> / <code>ADMIN</code> <code>PASSWORD</code> / <code>AUTH_SECRET</code> not set. Auth fails closed by design.
First turn of a call is slow, later turns fine	The agent subprocess spawning. Expected; a paid plan that never idles reduces it.